

Unit 5 - Solving Equations

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The following B.E.S.T. Standards will be covered in this unit:

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

- *Clarification 1: Instruction includes using formulas for temperature, perimeter, area and volume; using equations for linear (standard, slope-intercept, and point-slope forms), and quadratic (standard, factored, and vertex forms) functions.*
- *Clarification 2: Within the Mathematics for Data and Financial Literacy course, problem types focus on money and business.*

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

- *Clarification 1: Instruction focuses on analyzing a given function that models a real-world situation and writing constraints that are represented as linear equations or linear inequalities.*

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

- *Clarification 1: Within this benchmark, the expectation is to solve systems using elimination, substitution and graphing.*
- *Clarification 2: Within the Algebra 1 course, the system is limited to two equations.*

Unit 5, Lesson 1: Solving Multi-Step Equations



Benchmark

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.



Learning Target

☐ I can solve multi-step linear equations.



5.1.1 Warm-Up: Aviation Careers

Aviation careers include pilots, air-traffic controllers, airport operators, airfield maintenance, aircraft rescue and firefighting, law enforcement and security officers, and project management.



1. Which of these careers in aviation had you not heard of before?
2. Have you ever thought about a career in aviation?



5.1.2 Refresher: Solving Two-Step Equations

1. Work with your partner to complete the statements using the words from the bank. Words may be used more than once.

coefficient

constant

variable

- a. In the expression $7x - 2$, x is the _____,
 7 is a _____, and -2 is a
_____.
 - b. When solving an equation, the goal is to isolate the
_____.
2. Solve each equation and verify your solution using substitution.
 - a. $8.25 - 1.5y = 1.8$
 - b. $21 = 36\left(\frac{2}{3}a + \frac{5}{4}\right)$

3. Compare your work with your partner's. Summarize any similarities and/or differences in your solution methods.



5.1.3 Collaboration: Solving Multi-Step Equations

Javier and Arnav solved the same two equations, but they used different first steps.

1. In the table, each student's first step for solving each equation is started in blue. Work with your partner to finish solving the equations, continuing from each student's first step.

Javier	Arnav
$47 - 5.7m = 2.3m + 52$ $+ 5.7m \quad + 5.7m$	$47 - 5.7m = 2.3m + 52$ $-47 \qquad \qquad -47$
$-\frac{1}{4}(28p + 60) = -6p + 1$ $(-\frac{1}{4})(28p) + (-\frac{1}{4})(60) = -6p + 1$	$\frac{-\frac{1}{4}(28p+60)}{-\frac{1}{4}} = \frac{-6p+1}{-\frac{1}{4}}$

2. Discuss with your partner why both students had the same solutions for each equation, even though they used different strategies.

3. Discuss which strategies you preferred to start with when solving each of these equations. Summarize your discussion.



5.1.4 Guided Practice: Solving Multi-Step Equations

1. Solve each equation and verify the solution using substitution.

a. $8 + 5(16 + 9g) = 35g$

b. $49 - 6d = \frac{3}{2}(28d - 10)$

c. $\frac{62k-27}{11} = 8k + 7$



5.1.5 Your Turn!

1. Solve each equation.

a. $4.7y + 1.6 - 0.2y = 6.6$

b. $\frac{8}{3} + 29z + 1 = 7z - \frac{1}{3}$



5.1.6 Wrap-Up: Cool Down

1. Solve each equation and check the solution using substitution.

a. $23.5 - 32p = 16.5p + 4.1$

b. $\frac{3}{4}(12n - 10) = 19n - 3.5$



Unit 5, Lesson 2: Solving Equations and Rearranging Formulas



Benchmarks

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.



Learning Targets

- ☐ I can solve multi-step linear equations.
- ☐ I can isolate a quantity of interest in a formula by rearranging a formula.



5.2.1 Warm-Up: Closest to Zero

- Place a digit in each blue box to make the equation true for only one value of x . Use only the digits 1 to 9, but digits can be used more than once.

$$\boxed{}x + \boxed{} = \boxed{}x + \boxed{}$$



5.2.2 Collaboration: Find the Error

Sandra, Jerry, Marly, and Xavier solved the same equation and each found different solutions. Each student's work and solution are shown in the table.

- Work with your partner to examine the work shown and determine who solved the equation correctly. Then, for the students who made errors, circle the step where the error was made and explain the error.

Name	Student's Work and Solution	Is the solution correct?
Sandra	$9 - \frac{3}{4}(16 - 2w) = \frac{1}{3}w + 11$ $\frac{3}{4}(16 - 2w) = \frac{1}{3}w + 2$ $12 - \frac{3}{2}w = \frac{1}{3}w + 2$ $10 - \frac{3}{2}w = \frac{1}{3}w$ $10 = \frac{11}{6}w$ $\frac{60}{11} = w$	☐ Yes ☐ No
	Error Explanation	
Jerry	$9 - \frac{3}{4}(16 - 2w) = \frac{1}{3}w + 11$ $9 - 12 + \frac{3}{2}w = \frac{1}{3}w + 11$ $-3 + \frac{3}{2}w = \frac{1}{3}w + 11$ $-3 = \frac{7}{6}w + 11$ $-14 = \frac{7}{6}w$ $-12 = w$	☐ Yes ☐ No
	Error Explanation	

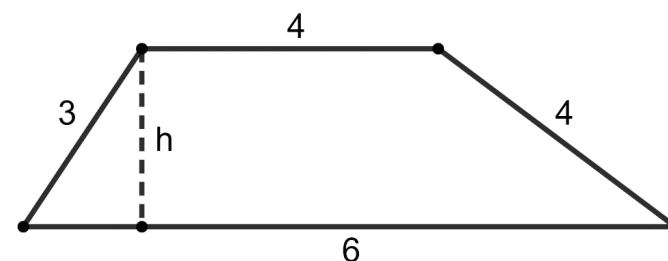
Name	Student's Work and Solution	Is the solution correct?
Marly	$9 - \frac{3}{4}(16 - 2w) = \frac{1}{3}w + 11$ $9 - 12 + \frac{3}{2}w = \frac{1}{3}w + 11$ $-3 + \frac{3}{2}w = \frac{1}{3}w + 11$ $\frac{3}{2}w = \frac{1}{3}w + 14$ $\frac{7}{6}w = 14$ $w = 12$	☐ Yes ☐ No
	Error Explanation	
Xavier	$9 - \frac{3}{4}(16 - 2w) = \frac{1}{3}w + 11$ $9 - 12 - \frac{3}{2}w = \frac{1}{3}w + 11$ $-3 - \frac{3}{2}w = \frac{1}{3}w + 11$ $-\frac{3}{2}w = \frac{1}{3}w + 14$ $-\frac{5}{6}w = 14$ $w = -16.8$	☐ Yes ☐ No
	Error Explanation	



5.2.3 Guided Instruction: Rearranging Formulas

The formula for the area of a trapezoid is $A = \frac{1}{2}(b_1 + b_2)h$, where b_1 and b_2 are the lengths of the bases and h is the height of the trapezoid.

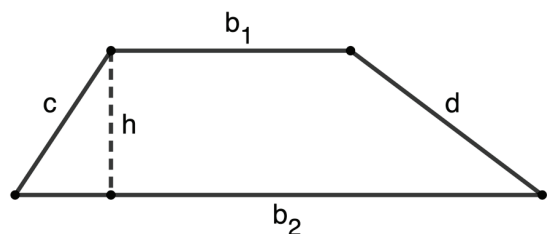
Stacey knows the area of the trapezoid shown is 12.5 square units, and uses the values given to determine the height, h , of the trapezoid. Her work is shown.



- Complete the table to explain each step of Stacey's work.

Stacey's Work	Explanation of Steps
$A = \frac{1}{2}(b_1 + b_2)h$	Formula for area of a trapezoid
$12.5 = \frac{1}{2}(4 + 6)h$	Substitute known values
$25 = (4 + 6)h$	
$\frac{25}{(4 + 6)} = h$	
$\frac{25}{(10)} = h$	
$2.5 = h$	

2. Stacey wants to rearrange the area formula for a trapezoid to isolate the variable h before replacing the values of the other variables.



- a. Use Stacey's work on the previous problem to help solve the equation for h .

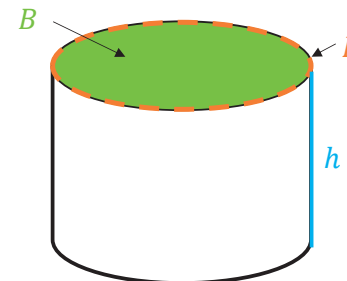
$$A = \frac{1}{2}(b_1 + b_2)h$$

- b. Check your work by substituting the values from the previous question. If correct, $h = 2.5$.

- c. Explain the similarities and differences between rearranging the formula to isolate h and Stacey's work to find the value of h .

Similarities	Differences

3. The formula for the total surface area of a cylinder, S , is $S = 2B + Ph$, where B is the area of the base, P is the perimeter of the base, and h is the height of the cylinder.



Rearrange the formula for the surface area of a cylinder to isolate the variable h .

$$S = 2B + Ph$$



5.2.4 Your Turn!

Rearrange the formulas to isolate the specified variables.

1. The formula for force, F , is mass, m , times acceleration, a , or $F = ma$. Solve for mass, m .
2. The formula $C = 2\pi r$ is used to find the circumference, C , of a circle with radius r . Solve the equation for the radius, r .
3. The formula to convert degrees Fahrenheit, F , into degrees Celsius, C , is $C = \frac{5}{9}(F - 32)$. Solve for F .



5.2.5 Wrap-Up: Growth Mindset

1. Choose **two** of the following prompts to complete.
 - a. Describe a new strategy you learned today.
 - b. Describe a mistake you made and what you learned from it.
 - c. Explain one way you challenged yourself today.

Unit 5, Lesson 3: Interpreting Equations



Benchmarks

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can define variables in a real-world context.
- ☐ I can interpret equations and their parts in a given context.



5.3.1 Warm-Up: Orange Juice

1. A typical orange produces 86 grams of juice. The juice contains 9 grams of carbohydrates.
 - a. If 1 gram of carbohydrate contains 4 calories, how many calories come from the carbohydrates in one orange?
 - b. The total number of calories of the juice from one orange is 39. What percentage of those calories comes from the carbohydrates in the juice?
 - c. Carbohydrates include sugar. The sugar that occurs in orange juice is natural sugar. If a typical orange contains 7 grams of sugar, what percentage of the calories from carbohydrates comes from sugar? Round to the nearest tenth.



5.3.2 Guided Instruction: What is a Parameter?

1. In the expression $4x$, the 4 is the coefficient and the x is the variable. The expression $4x$ could be used to represent the number of calories in orange juice that comes from carbohydrates since 1 gram of carbohydrate contains 4 calories.



A **parameter** is a quantity that influences the output.

In the expression $4x$, what is the parameter?

2. The cost of an item is \$8.50.
 - a. Write an expression that can be used to determine the cost of x items.
 - b. Discuss with your partner why \$8.50 is a parameter. Summarize your discussion.

3. The price of an item is P . During a sale, the price is discounted by 20%, so the new price is $0.8P$. This could also be expressed as $(1 - 0.20)P$.

Complete the table.

Expression	Parameter
$0.8P$	
$(1 - 0.20)P$	

4. Discuss with your partner which values in the U-Pick Blueberries sign are parameters. Justify your choice.

U-Pick Blueberries

\$4 per pound

\$1 parking



5. The average price, P , in dollars, of a 16 ounce bag of potato chips for each year, x , since 1980, can be modeled by the function $P(x) = 0.07x + 2.20$.
 - a. Which value in the function is a parameter?

b. What does the value 2.20 mean in terms of the price of potato chips?

c. What does the value 0.07 mean in terms of the price of potato chips?

6. The students in the Visual and Performing Arts program at Booker High School in Sarasota County, Florida are planning a field trip to the Ringling Art Museum. The cost of admission is \$5 and the cost of renting a bus is \$115. There will be 9 chaperones. The budget for the field trip is \$600.

a. Work with your partner to determine which values, if any, are parameters. Write a justification for each.

Value	Parameter?	Justification
5	☐ yes ☐ no	
9	☐ yes ☐ no	

115	☐ yes ☐ no	
600	☐ yes ☐ no	

b. Use the values to complete the equation that can be used to find the number of students, x , who can go on the field trip.

$$\underline{\hspace{1cm}}(x + \underline{\hspace{1cm}}) + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$



5.3.3 Activity: Card Sort

- Work with your partner to sort the cards provided by your teacher.
 - Record the two lowercase letters on the card that matches the meaning of each value.

Real-World Context	Meaning of Each Value	
1	1991	
	6.88	
	0.37	
	3	
2	2010	
	19.116	
	1.352	
	43.452	
3	8	
	0.28	
	34.60	
	2.40	

- Record the letter and the equation that matches the context for each scenario.

Real-World Context	Equation
1	
2	
3	



5.3.4 Wrap-Up: Growth Mindset

- Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

My favorite mistake from today was ...

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Unit 5, Lesson 4: Writing and Solving Real-World Equations



Benchmark

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.



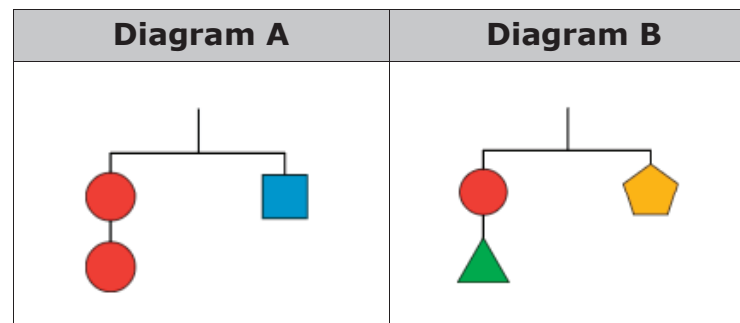
Learning Target

- ☐ I can write and solve two-step equations given a real-world context.

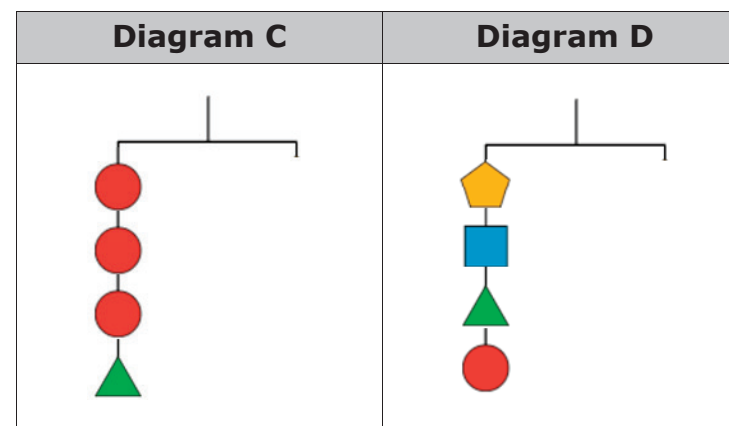


5.4.1 Warm-Up: Hanger Diagrams

Two hanger diagrams are shown. They are balanced with an equal weight hanging from both sides.



- Use the diagrams above to complete the right side of each diagram so that Diagrams C and D are balanced. You cannot copy the left side of the diagram.



2. James drew a pentagon and a square for the right side of Diagram C. Draw pictures or explain how James could conclude that a pentagon and a square are equivalent to the left side of Diagram C.



5.4.2 Guided Instruction: Solving Real-World Problems by Writing and Solving Equations

1. The U.S. Census Bureau recorded a population of approximately 18.8 million people living in Florida in 2010. Since 2010, there has been an average increase of 0.3 million people per year.
 - a. What are the quantities that are changing in this scenario?
 - b. Define the variables a and p .
 - c. What is the parameter?
 - d. What is the constant?
 - e. Which equation represents the situation?
 - $p = 18.8a + 0.3$
 - $p = 0.3a + 18.8$
 - $a = 18.8p + 0.3$
 - $a = 0.3p + 18.8$
 - f. Use your equation to determine in what year Florida's population was approximately 21.2 million.

2. Simone purchased 4 skinny vanilla lattes for her and her three friends. A 6% sales tax was applied to the purchase, and she gave the rest of her money as a \$3.86 tip for a total of \$24.

- Define the variable in this context.
- Discuss with your partner what the parameters are in this context.
- Complete the table to write an equation that represents the context.

Total \$ Spent	=	# of Lattes	(Latte Price)	(1 + Tax Rate)	+	Tip
	=				+	

- Solve the equation for the unknown.
- Interpret the solution.



5.4.3 Guided Practice: Solving Real-World Problems by Writing and Solving Equations

1. The popularity of Twitter can be measured using the number of active accounts. In January of 2015, Twitter had approximately 302.5 million active accounts worldwide. Each year, the number of active accounts has increased by about 10.2 million.

- Define the variables in this context.
- What is the parameter?
- What is the constant?
- Which variable is the output in this context?

e. Write an equation to represent this situation.

f. Write a one-variable equation that can be used to determine the year Twitter will have 425 million active accounts if the trend continues.

g. Solve the equation from Part F.

h. Interpret the solution.

i. Are there any constraints to the given situation?



5.4.4 Try It: Solving Real-World Problems by Writing and Solving Equations

1. Since the fourth quarter of 2014, the flash storage capacity of smartphones has grown by approximately 12 gigabytes a year. The average flash storage in 2014 was 15.9 gigabytes.

a. Write an equation to represent this situation, defining both variables.

b. Assuming the trend continues, write a one-variable equation that can be used to find the average flash storage capacity of smartphones in 2025.

c. Solve your equation from Part B.

d. Interpret the solution.

e. Discuss any constraints of the situation with your partner.



5.4.5 Wrap-Up: Ticket Out the Door

1. James is going to play paintball. He is charged \$25.95 for equipment rental, plus \$6.57 per box of 100 paintballs. If James spent \$58.80, how many boxes of paintballs did he purchase?
 - a. Write and solve an equation representing this situation.
 - b. Interpret the solution.

Unit 5, Lesson 5: Equations and Solutions in the Real World



Benchmarks

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



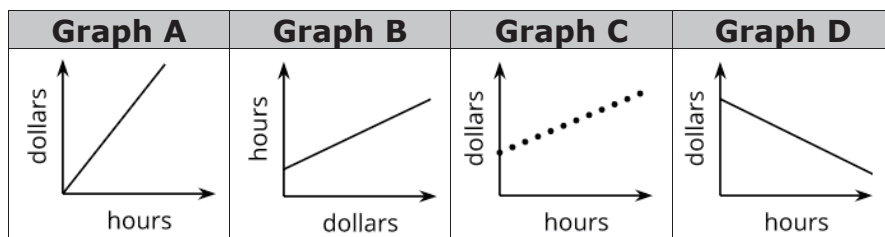
Learning Targets

- ☐ I can write and solve multi-step equations in a real-world context.
- ☐ I can interpret solutions as viable or non-viable in a given context.



5.5.1 Warm-Up: Which One Doesn't Belong?

Four graphs are shown.



- Complete the statement to explain which graph does not belong.

- ☐ Graph A
- ☐ Graph B
- ☐ Graph C
- ☐ Graph D

does not belong because _____

_____.

- Find a classmate who selected a different graph that does not belong. Write their explanation in the table. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary of their explanation stating that your summary is correct.

My Summary of _____'s Description	Initials



5.5.2 Guided Instruction: Writing and Solving Equations Representing Real-World Contexts

- The school choir is taking a day trip to Volcano Bay. They got a special ticket rate of \$65 per person, but the cost of the bus is \$639.90. If they fill every seat on the bus, then each person will pay a total of \$76.85 for bus transportation and an admission ticket.
 - Complete the table to write two different algebraic expressions for the cost of this trip, where s represents the number of people on the trip.

Expression A	Expression B
<i>ticket rate per person plus the total bus cost</i>	<i>total amount paid per person, if all seats are full</i>

If all the seats on the bus are full, then Expression A and Expression B will be equivalent to each other. Therefore, they can be set equal to each other to create an equation.

- Write and solve this equation.



c. Interpret the solution in context.

d. Discuss with your partner why 52.7 would not be a viable solution for this context. Summarize your discussion.

2. Max and his friends are planning a chartered fishing trip off the coast of Fort Lauderdale. Fishing Adventures charges a base fee of \$299.22, plus \$21.90 per person for a 4-hour trip. However, Max found a 10% off coupon for Ocean Excursions. Normally Ocean Excursions offers 6-hour trips for \$50.58 per person plus a flat fee of \$70 for equipment. With the 10% off coupon, the two options would cost Max and his friends the same amount of money.

a. Define a variable to represent the unknown.

b. Write and solve an equation representing this situation.

c. Interpret the solution.



5.5.3 Guided Practice: Solving Real-World Problems by Writing and Solving Equations

1. Marisol was surprised when she discovered last week that she had worked the same number of hours and made the same amount of money by working two different jobs. As a server at The Main Street Grill, she earned \$5.85 per hour plus \$74.40 in tips. Working at Fresh Fashions she earned \$8.65 per hour plus \$22.60 in sales bonuses.
 - a. Define a variable to represent the unknown.
 - b. Write an expression that represents the amount Marisol earned working at The Main Street Grill.
 - c. Write an expression that represents the amount Marisol earned working at Fresh Fashions.
 - d. Write the equation that represents last week, when she earned the same amount of money working the same number of hours at both jobs.
 - e. Solve the equation.
 - f. Interpret the solution.
 - g. Explain whether the mathematical solution to the equation is viable in this context.
 - h. Complete the statement. Use the space provided to show any work.

Last week, Marisol worked a total of _____ hours and earned a total of \$_____.



2. The Zamora family of four is flying from Orlando to New York's JFK airport. They found tickets with Sky Airlines for \$74 per person round trip, but there is a \$25 charge per bag. Mrs. Zamora found that for the total number of bags they plan to bring, they can save money by upgrading their tickets. With the upgrade, they would pay \$103 per person for the tickets, \$5 per bag, and they would still save \$4 total.

- Define a variable to represent the unknown.
- Write and solve an equation that represents this situation.
- Interpret the solution.



5.5.4 Your Turn!

1. The student council at Cypress Creek High School had a jog-a-thon fundraiser with a designated pledge amount per mile. Each class can raise additional money through donations of any amount. The freshman and sophomore classes raised the same amount of money. The freshman class ran a total of 562 miles and collected \$131.25 in donations. The sophomore class collected \$185 in donations and ran a total of 519 miles.

- Define a variable to represent the unknown.
- Write and solve an equation that represents this situation.
- Interpret the solution.
- How much money did each class raise?



5.5.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.



I need help.



I got it.



I could help someone.

Identifying and defining the variable for a real-world context			
Writing a multi-step equation to represent a context			
Solving multi-step one-variable linear equations			
Interpreting the meaning of the solution in a given context			



Unit 5, Lesson 6: Exploring Solutions to Systems of Equations



Benchmark

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.



Learning Target

- ☐ I can relate the solution of a system of equations to its graph and equations.



5.6.1 Warm-Up: Mindsets in Math

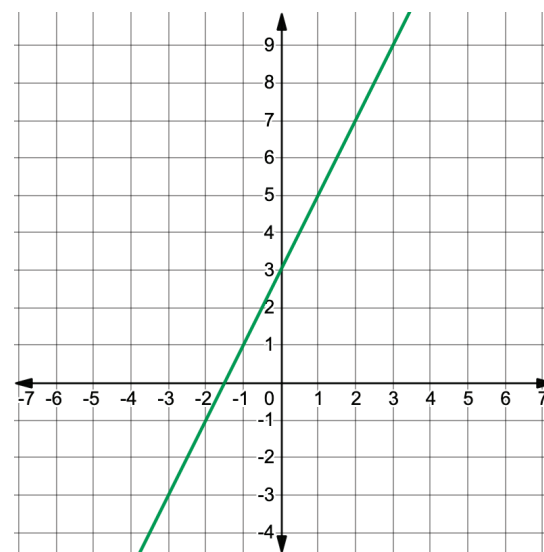
1. Some people believe the myth that math intelligence is something people are only born with. This is called having a “fixed mindset.” Others recognize that math intelligence is something anyone can develop, which is called having a “growth mindset.”

- a. Describe something you have learned in math this year that you didn’t initially understand, but you do now.
- b. What helped you learn it?
- c. How does your mindset regarding math intelligence impact the way you learn and face challenges in math class?



5.6.2 Exploration: Solutions to Systems of Equations

1. The graph of the equation $y = 2x + 3$ is shown.

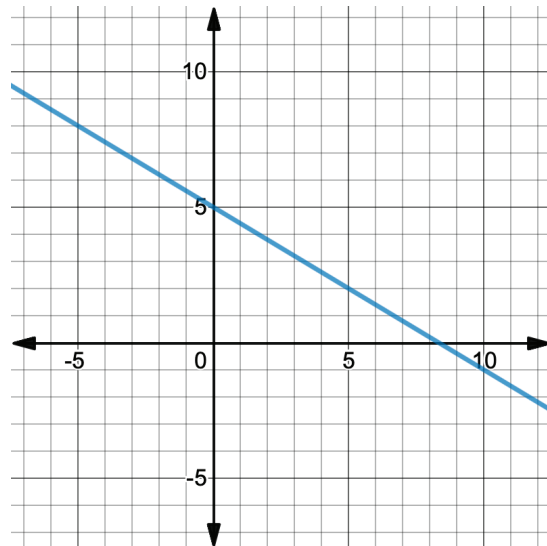


- a. Mark the point $(2, 7)$ on the graph. What do you notice?
- b. Complete the statement.
Since the point $(2, 7)$ lies on the line of $y = 2x + 3$, it is considered a(n)

intercept
solution
vertex

 to the equation.
- c. Verify algebraically that point $(2, 7)$ is a solution to the equation $y = 2x + 3$.

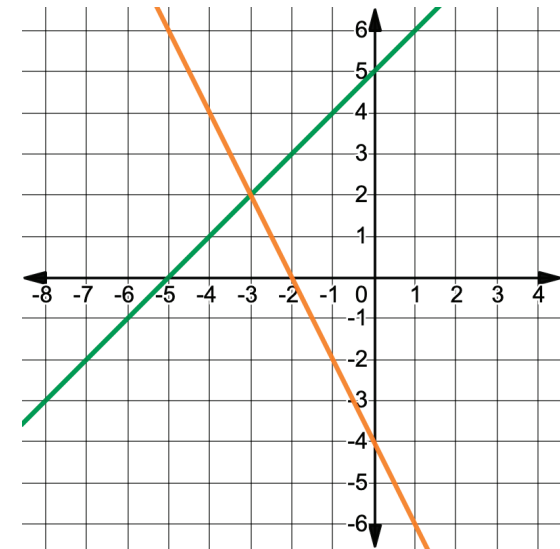
2. The graph of the equation $y = -0.6x + 5$ is shown.



Work with your partner to find two different solutions to the equation $y = -0.6x + 5$. Then, verify algebraically that each is a solution.

Solution 1: (____, ____)	Solution 2: (____, ____)
Algebraic Verification	Algebraic Verification

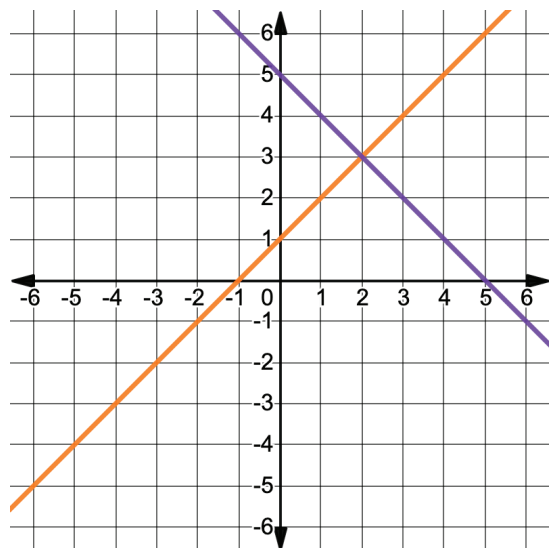
3. The graphs of two linear equations are shown.



- Discuss with your partner whether or not there is a common solution for both linear equations shown on the graph.
- Complete the statement.
The point (____, ____) lies on both lines, so it is considered a solution of both linear equations.

A **system of equations** is a set of two or more equations. Each equation contains two or more variables. The solution is all the values for the variables that make all equations in the system true.

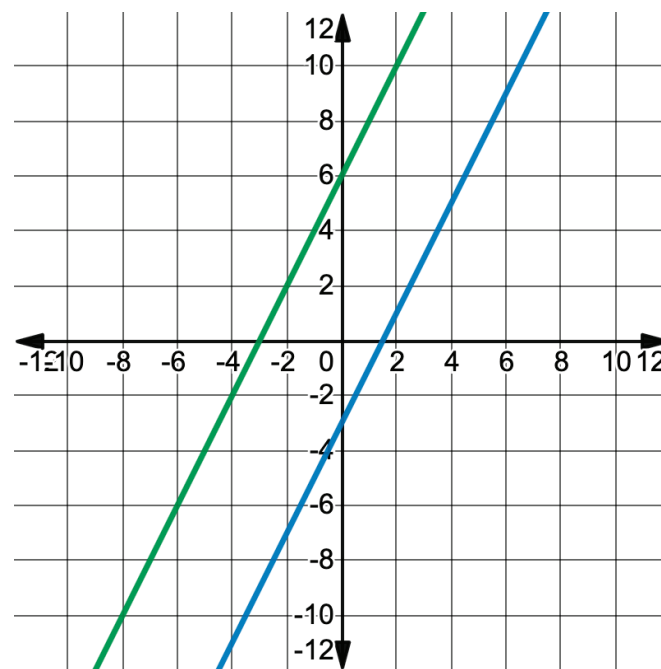
4. The graph of the system of equations, $\begin{cases} y = x + 1 \\ y = -x + 5 \end{cases}$, is shown.



- Work with your partner to determine the solution to the system of equations from the graph.
- Verify algebraically that your solution satisfies both equations in the system.

$y = x + 1$	$y = -x + 5$

5. The graph of a system of equations is shown.

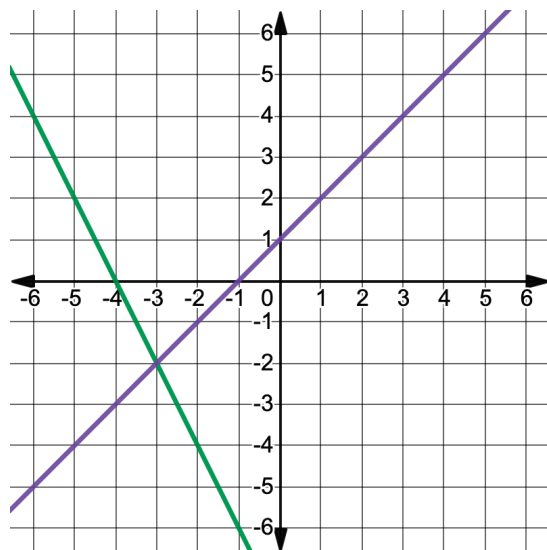


Work with your partner to describe the solution of the system.



5.6.3 Exploration: Equivalent Systems of Equations

1. Chloe, Armin, and Olivia are all solving the system of equations, $\begin{cases} y = -2x - 8 \\ y = x + 1 \end{cases}$. The graph of the system is shown.



- a. What is the solution to this system?

Each student rewrote the original system of equations in different ways. Their new systems and how they generated them are shown in the table.

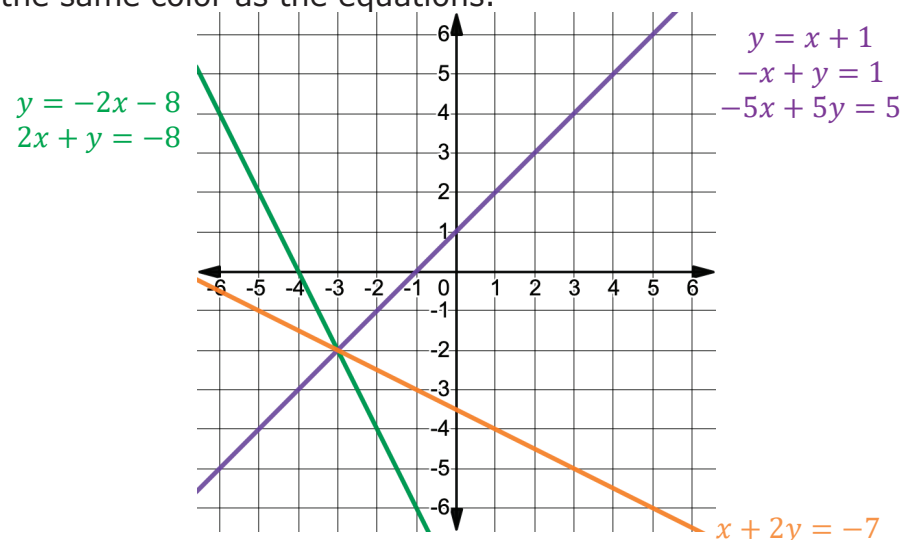
Student	Student's Work	New System
Chloe	$\begin{array}{r} y = -2x - 8 \\ +2x \quad +2x \\ \hline 2x + y = -8 \end{array}$ $\begin{array}{r} y = x + 1 \\ -x \quad -x \\ \hline -x + y = 1 \end{array}$	$\begin{cases} 2x + y = -8 \\ -x + y = 1 \end{cases}$
Armin	$\begin{array}{r} y = -2x - 8 \\ + (y = x + 1) \\ \hline 2y = -x - 7 \end{array}$ $\begin{array}{r} +x \quad +x \\ \hline x + 2y = -7 \end{array}$ $\begin{array}{r} y = x + 1 \\ -x \quad -x \\ \hline -x + y = 1 \end{array}$	$\begin{cases} x + 2y = -7 \\ -x + y = 1 \end{cases}$
Olivia	$\begin{array}{r} y = -2x - 8 \\ +2x \quad +2x \\ \hline 2x + y = -8 \end{array}$ $\begin{array}{r} 5(y = x + 1) \\ 5y = 5x + 5 \\ -5x \quad -5x \\ \hline -5x + 5y = 5 \end{array}$	$\begin{cases} 2x + y = -8 \\ -5x + 5y = 5 \end{cases}$

- b. Chloe rearranged her equations. Explain what Armin did to generate his new system.

c. Explain how Olivia generated her new system.

d. With your partner, discuss whose new system you think will result in the same solution as the original system. Then, summarize your discussion.

The graphs of the original system of equations and the newly created systems are shown on the same coordinate grid. The graphs that represent each equation are shown in the same color as the equations.



The graphs of $y = x + 1$, $-x + y = 1$, and $-5x + 5y = 5$ are on the same line, shown in purple on the coordinate grid. Therefore, these equations are **equivalent**, not equal. To be equal, the equations would be exactly the same.



When two entities have the same result, they are **equivalent**. Systems of equations with the same solution are **equivalent systems**.

e. Complete the statement.

Chloe, Armin, and Olivia all generated systems of

equations ☐ equal ☐ equivalent to the original system

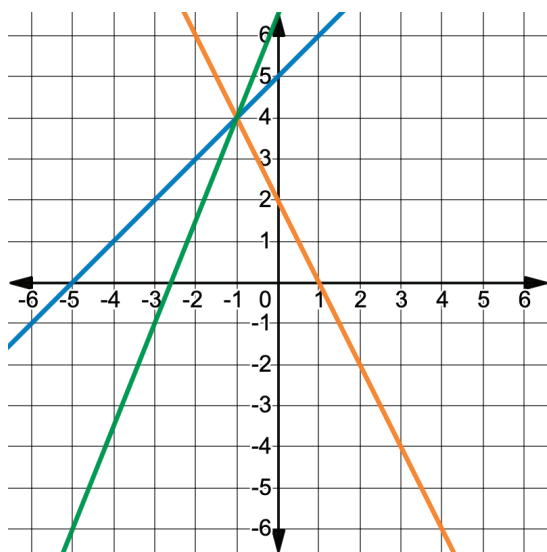
because all of the systems result in ☐ the same ☐ different

solution(s).



5.6.4 Bring it Together: Systems of Equations

1. The graphs of the systems of equations $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ and $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ are shown on the same coordinate plane.



Complete the statements.

- a. The solution of any system of equations is the

- ☐ point(s) of intersection.
- ☐ slope.
- ☐ y-intercept.

- b. The solution of the system $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ is (____, ____).

- c. The solution of the system $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ is (____, ____).

- d. The systems of equations $\begin{cases} 2x + y = 2 \\ x - y = -5 \end{cases}$ and $\begin{cases} 5x - 2y = -13 \\ x - y = -5 \end{cases}$ are _____ systems because they have the _____ solution.

2. Verify algebraically that the solution satisfies all of the equations included in either system.

$2x + y = 2$	$x - y = -5$	$5x - 2y = -13$



5.6.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Explorations that I knew were ...	Something I wondered during the Explorations was ...	Something I learned during the Explorations was ...

Unit 5, Lesson 7: Solving Systems of Equations by Graphing



Benchmarks

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can solve systems of equations by graphing, while noting constraints for a real-world context.
- ☐ I can interpret the solution to a system of equations in a real-world context.
- ☐ I can interpret the coefficients of a given system representing a real-world context.



5.7.1 Warm-Up: Solve the Puzzle

1. Determine the value of each figure so that the sum of each row and column match the total value shown.

				172
				136
				136
108	144	108	84	

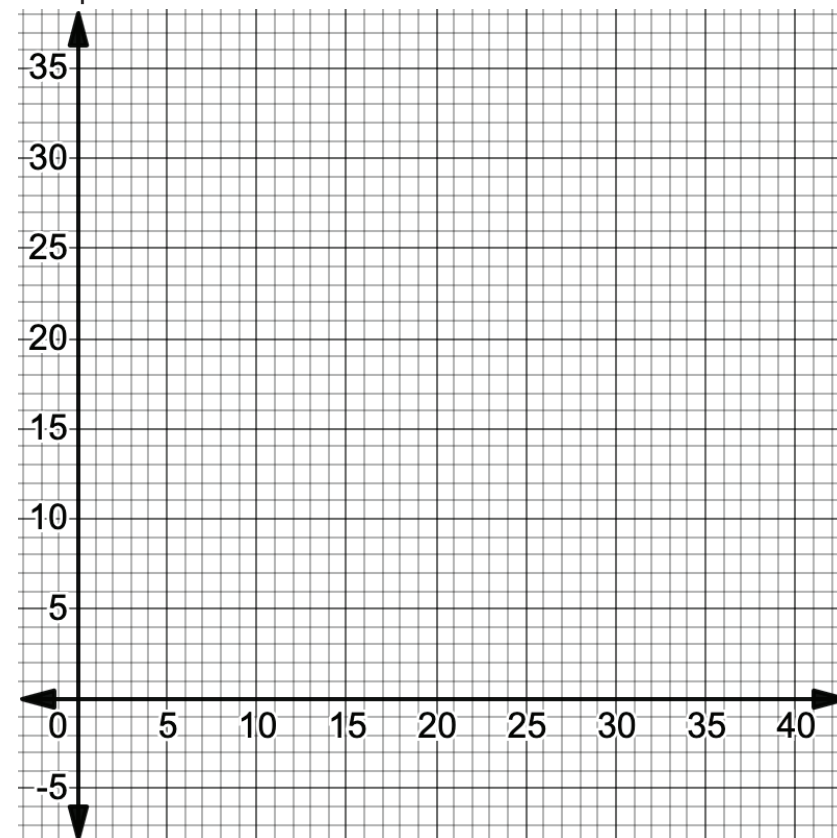
2. To solve the puzzle, explain which figure you started with and how you were able to find its value.
3. How did you find the values of the other figures after determining the value of the first figure?



5.7.2 Collaboration: Solving Systems of Equations by Graphing

1. Use the system of equations $\begin{cases} y = 0.6x + 5 \\ y = 0.3x + 14 \end{cases}$ to complete the following.

- a. Graph the system of equations on the coordinate plane.



- b. Complete the statement.

The point (____, ____) is the solution to the system.

2. The system of equations from the previous question can be used to represent the following situation.

Many businesses use Google to track trends to help them make business decisions. A value of 100 represents peak popularity of a search term. A marketing analyst found the following search trends when comparing interest in pho and ramen through search trends in the U.S. since 2010.

- In January 2010, pho had a search interest value of 14, which increased by an average of 0.3 each month until March of 2021.
 - In January 2010, ramen had a search interest value of 5, which increased by an average of 0.6 each month until March of 2021.
- a. Match each search term to the equation that could be used to model its trend from January 2010 to March 2021.

Search Term	Equation
pho	○ $y = 0.6x + 5$ ○ $y = 0.3x + 14$
ramen	○ $y = 0.6x + 5$ ○ $y = 0.3x + 14$

- b. Discuss with your partner what the solution to the system found in the previous question means in this context. Then, summarize your discussion.

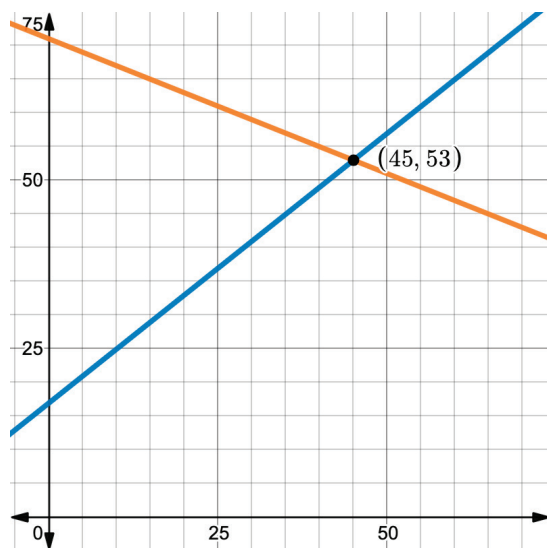
- c. Work with your partner to explain how negative x values relate to the situation.

- d. Discuss with your partner any possible constraints on the x -values in this context.



5.7.3 Guided Instruction: Solving Systems of Equations by Graphing and Interpreting Solutions

1. The graph represents the trends in the percentage of U.S. adults who read e-books versus print books each year since 2011.



- a. Let s represent the number of years since 2011 and p represent the total number of people using the reading format. Match the situation with the appropriate equation and line from the system shown on the graph.

Situation	Equation	Graph
In 2011, 17 percent of the U.S. adults surveyed read e-books, and the percentage increased $\frac{4}{5}\%$ each year after.	☐ $p = -0.4s + 71$ ☐ $p = 0.8s + 17$	☐ Orange line ☐ Blue line

In 2011, 71 percent of the U.S. adults surveyed read print books, and the percentage decreased $\frac{2}{5}\%$ each year after.

- ☐ $p = -0.4s + 71$
☐ $p = 0.8s + 17$

- ☐ Orange line
☐ Blue line

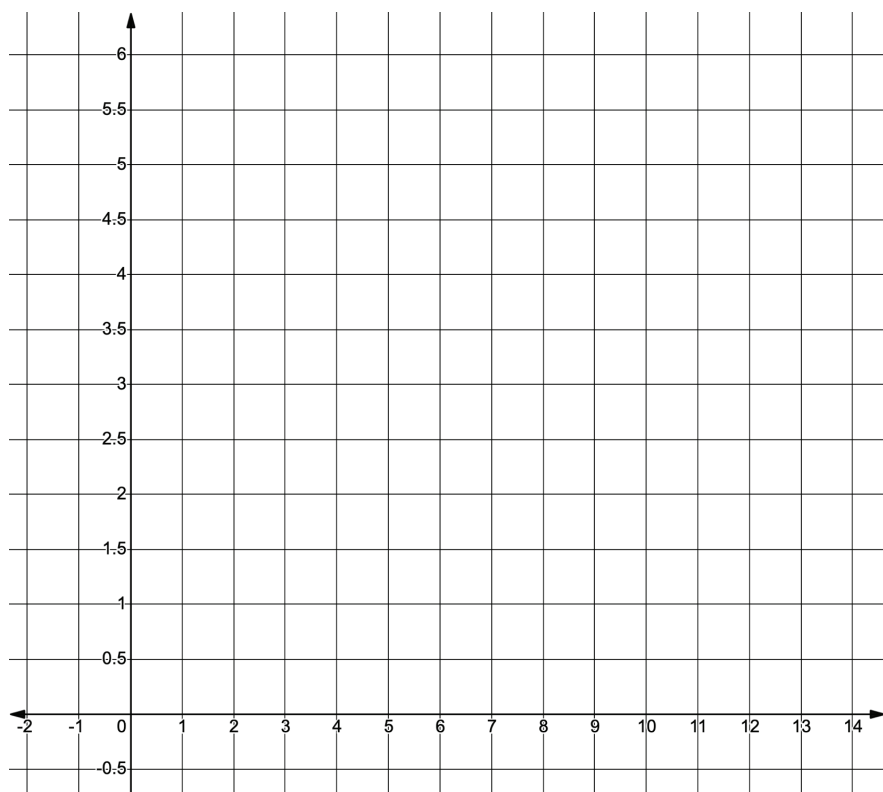
- b. If the trends continue, what year will e-books and print books have the same number of readers?
- c. What percentage of the U.S. adult population is predicted to be using each reading format at that time?
- d. The variable s represents the years since 2011. If $s = -2$, what year is being represented?
- e. Discuss with your partner any real-world constraints that may impact these trends.



5.7.4 Guided Practice: Solving Systems by Graphing and Interpreting Solutions

1. Levi is comparing the average daily precipitation in millimeters (mm) of Calhoun County, Florida to the entire state of Florida. Based on the data he collected from 1999 to 2010, Levi uses the equation $a = \frac{1}{5}w + 3.9$ to represent the precipitation in Calhoun County, FL, and $a = \frac{1}{5}w + 3.4$ to represent the precipitation in the state of Florida. In both equations in the system $\begin{cases} a = \frac{1}{5}w + 3.9 \\ a = \frac{1}{5}w + 3.4 \end{cases}$, a represents the average daily rainfall, in mm, and w represents the number of years since 1999.

a. Graph the system of equations.



- b. What does the slope of the lines represent in the given context?
- c. What was the total average rainfall for Calhoun County in 2003?
- d. Explain whether or not Calhoun County typically has more or less precipitation than the state.

e. Complete the statements.

The two equations in the system are ☐ intersecting ☐ parallel

lines with ☐ the same slope, ☐ different slopes, so they ☐ will ☐ will not

intersect. Therefore, there ☐ is ☐ is not a year when

Calhoun County had the same daily average

precipitation as the state of Florida. There

- ☐ is one solution
☐ is no solution
☐ are infinitely many solutions

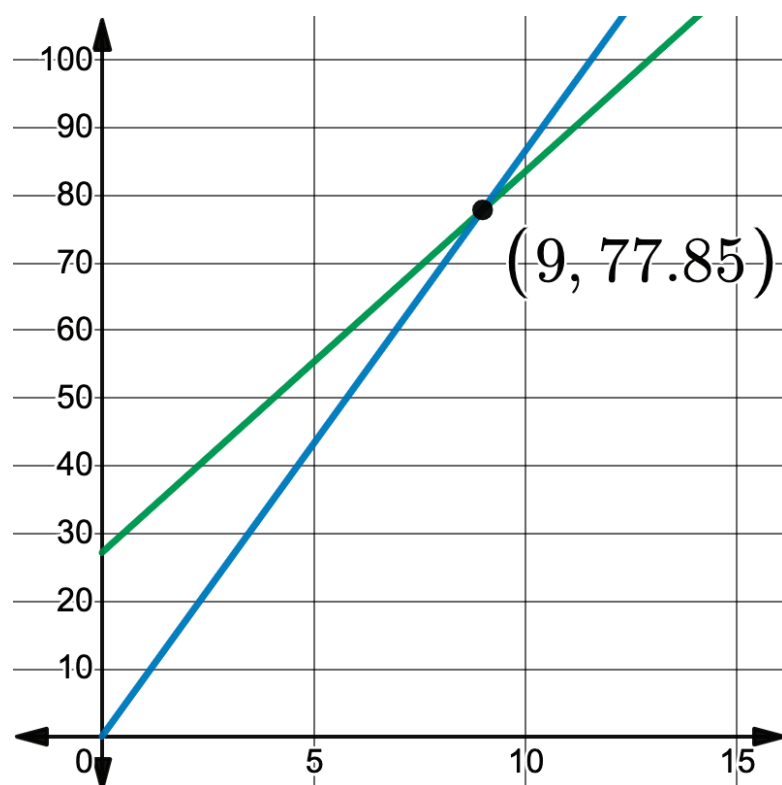
to this system.



5.7.5 Try It: Solving Systems by Graphing and Interpreting Solutions

1. Remy just started working at the local salon. She is paid \$5.63 per hour and earns, on average, \$27.18 dollars in tips each day. Suki just started working at Publix. Since she does not receive tips, minimum wage for her is \$8.65 an hour. The system of equations $\begin{cases} m = 5.63h + 27.18 \\ m = 8.65h \end{cases}$ can be used to represent the amount of money earned, m , each day for working h hours.

The graph of the system is shown.



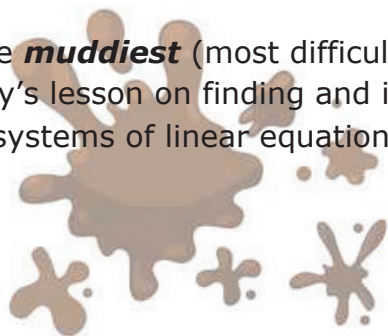
- a. Discuss with your partner why the graphs do not include negative h and m values.

- b. Label each line with the name of the person whose earnings it represents.
- c. What is the solution to this system of equations?
- d. Interpret the meaning of the solution in the context.
- e. Discuss with your partner the circumstances under which either Remy or Suki would earn more money per day. Then, summarize your discussion.



5.7.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on finding and interpreting solutions to systems of linear equations using graphs?



Unit 5, Lesson 8: Solving Systems of Equations by Substitution



Benchmarks

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



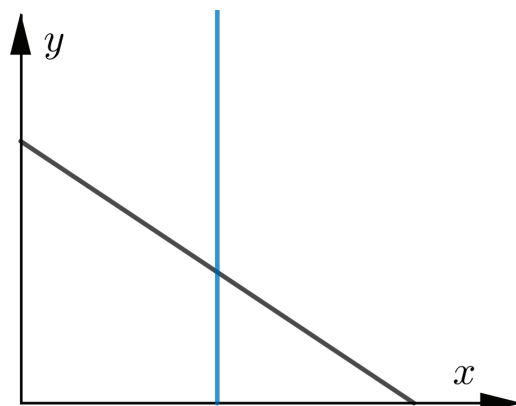
Learning Targets

- ☐ I can solve systems of equations by substitution.
- ☐ I can write a system of equations from a real-world context and solve by substitution.
- ☐ I can interpret the solution to a system of equations in a real-world context.



5.8.1 Warm-Up: Math Talk

- The graph of a system of linear equations is shown.



For each system given in the table, determine whether or not the system could be represented by the graph and explain how you know.

System	Could the graph represent this system?	Explain
$\begin{cases} x + 2y = 8 \\ x = -5 \end{cases}$	☐ yes ☐ no	
$\begin{cases} y = -7x + 13 \\ y = -1 \end{cases}$	☐ yes ☐ no	
$\begin{cases} 3x = 8 \\ 3x + y = 15 \end{cases}$	☐ yes ☐ no	
$\begin{cases} y = 2x - 7 \\ 4 + y = 12 \end{cases}$	☐ yes ☐ no	



5.8.2 Guided Instruction: Writing and Solving Systems of Equations Using Substitution

While systems of equations can be solved by graphing, they can also be solved algebraically. One method for solving systems of equations algebraically is **substitution**.



The **substitution property of equality** states that if $a = b$, then b may be substituted for a in any expression containing a .

- Use the system of equations $\begin{cases} y = 2x - 7 \\ x - 3y = 12 \end{cases}$ to complete the following.
 - Victor notices from the first equation of the system that the value of y is equal to $2x - 7$, so he substitutes this expression for y in the second equation of the system, as shown.

$$x - 3(2x - 7) = 12$$

Discuss with your partner how Victor's thought process applies the substitution property of equality.
 - Solve the equation Victor created using substitution.

$$x - 3(2x - 7) = 12$$

- c. The solution to a system of equations is an ordered pair. Victor's equation was used to find the x -value of the solution. Substitute the x -value into either equation in the system to find the y -value of the solution.

- d. What is the solution to the system $\begin{cases} y = 2x - 7 \\ x - 3y = 12 \end{cases}$?

2. In 1983, the composition of pennies in the United States was changed due, in part, to the rising cost of copper. Pennies minted after 1983 weigh 2.50 grams (g), while the earlier copper pennies weigh 3.11 g. A roll of pennies contains 50 coins. Suppose a particular roll of pennies weighs 132.32 g.

A system of equations can be written to represent this context. To write a system of equations, consider the unknown variables of interest and the information given.

- a. What information is given about the roll of pennies?
- b. Write a two-equation system of equations representing this information, where x represents the number of pennies minted prior to 1983 in the roll and y represents the number of pennies minted after 1983 in the roll.

- c. Solve the system of equations using substitution.

- d. Interpret the solution.



5.8.3 Guided Practice: Writing and Solving Systems of Equations Using Substitution

1. Solve the system of equations using substitution.

$$\begin{cases} x = 3y \\ \frac{4}{3}x - 7y = \frac{9}{2} \end{cases}$$

2. The marketing department at Sea World compared the number of visitors at their Orlando, Florida location with that of their San Diego, California location from 2008 to 2017 to help analyze marketing needs and make predictions.

In 2008, approximately 4.4 million people visited the San Diego location, while approximately 5.9 million people visited the Orlando location. Since then, the number of people visiting each location has been declining — San Diego by 0.1 million people per year and Orlando by 0.2 million people per year.

- a. Name two variables that are being related in this situation.

- b. Write a system of equations to represent the trends at each Sea World Park since 2008.

- c. Solve the system of equations using substitution.

- d. Interpret the solution.



5.8.4 Collaboration: Error Analysis

1. Feng is using substitution to solve the system of equations $\begin{cases} y = 3x - 4 \\ x = -y + 24 \end{cases}$. Feng's partial work is shown.

$$y = 3x - 4$$

$$x = -y + 24$$

$$x = -(3x - 4) + 24$$

$$x = -3x - 4 + 24$$

$$x = -3x + 20$$

$$4x = 20$$

$$x = 5$$

- a. Feng made an error solving for x . Circle the step where he made the error. Then correct his work in solving for x in the system in the space beside his work above.
- b. Using the corrected x -value, find the solution to the system. Show your work.

2. Micah solves the same system, but starts a different way.

Micah's first step is shown. Complete Micah's work to find the solution to the system.

$$y = 3x - 4 \qquad x = -y + 24$$

$$y = 3(-y + 24) - 4$$

3. Discuss with your partner the similarities and differences between Feng's and Micah's solution methods. Then, summarize your discussion in the table.

Similarities	Differences



5.8.5 Your Turn!

1. Cherie receives two different job offers after graduating.
- Job A offers Cherie a salaried position where she will earn \$35,000 total a year.
 - Job B offers Cherie an hourly position where she will earn \$24 per hour as well as an annual bonus of \$500.

The system of equations $\begin{cases} w = 35,000 \\ w = 24h + 500 \end{cases}$ represents this situation, where h is the number of hours Cherie works, and w is her total wages.

- a. Solve the system using substitution.

- b. Interpret the solution.



5.8.6 Wrap-Up: Ticket Out the Door

1. Solve the system of equations using substitution.

$$\begin{cases} 6x - 5y = -8 \\ y = -\frac{1}{2}x + 5 \end{cases}$$

Unit 5, Lesson 9: Solving Systems of Equations by Elimination



Benchmark

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.



Learning Target

- ☐ I can solve systems of equations by elimination.



5.9.1 Warm-Up: Goals and Aspirations

Nilo Cruz is a Cuban American playwright. Born in Matanzas, Cuba in 1960, his family immigrated to Miami, Florida in 1970. He first studied theater at Miami-Dade Community College before moving to New York City to begin his career. He later earned his Master's in Fine Arts from Brown University in 1994. Cruz served as the playwright-in-residence for the New Theater in Coral Gables, Florida in 2001. While there, he wrote *Anna in the Tropics*, a play he later won a Pulitzer Prize for in 2003. Nilo Cruz was the first Latino to win the Pulitzer Prize for Drama.



1. If you win a prize for a major accomplishment in the future, what would you want to be recognized for?



5.9.2 Guided Instruction: Solving Systems of Linear Equations using Elimination

In addition to solving systems of equations by graphing and using substitution, systems of equations can also be solved algebraically using **elimination**.

Solving a system by elimination means to *eliminate* one of the variables by adding or subtracting the system together.

1. Consider the system of equations $\begin{cases} 2x + 4y = 13 \\ -2x + 4y = 11 \end{cases}$

- a. Add the two equations together.

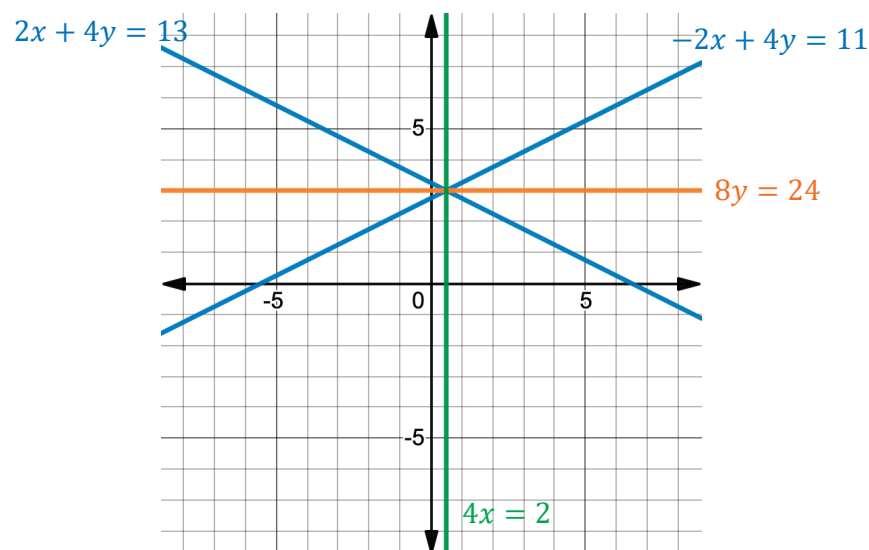
$$\begin{array}{r} 2x + 4y = 13 \\ +(-2x + 4y = 11) \\ \hline \end{array}$$

- b. Subtract one equation from the other.

$$\begin{array}{r} 2x + 4y = 13 \\ -(-2x + 4y = 11) \\ \hline \end{array}$$

- c. What do you notice about the resulting equations in Parts A and B?

- d. The graphs of the original equations in the system and the resulting equations from Parts A and B are shown on the same coordinate grid.



What do you notice?

- e. Use the resulting equation from Part A to solve the original system of equations algebraically.
- f. Verify your solution using the graph.



When equations in a system are added or subtracted, the resulting equation shares the solution of the original system.

2. Use the system of equations $\begin{cases} 5a - b = 9 \\ 2.5a + b = 6 \end{cases}$ to complete the following.

- a. Complete the statement.

- ☐ Adding
☐ Subtracting

the original equations in this

system would result in eliminating one of the

variables,

- ☐ a .
☐ b .

- b. Perform the operation selected in Part A.

$$\begin{array}{r} 5a - b = 9 \\ \underline{ (2.5a + b = 6)} \end{array}$$

- c. Solve the system using your equation from Part B.

$$\begin{array}{l} a = \underline{\hspace{2cm}} \\ b = \underline{\hspace{2cm}} \end{array}$$

- d. Riley wondered if it is possible to solve the system by eliminating the a -terms instead of the b -terms first. He suggested multiplying the second equation by 2 and then subtracting it from the first equation. The first steps Riley suggests are shown. Complete Riley's work and use the results to solve the system.

$$2(2.5a + b = 6)$$

$$\boxed{}a + \boxed{}b = \boxed{}$$

$$\begin{array}{r} 5a - b = 9 \\ -() \\ \hline \end{array}$$

$$a = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$



When solving a system of equations using **elimination**, one or both of the equations can be multiplied by a factor before adding or subtracting the equations. The resulting equation shares the same solution with the original system.



5.9.3 Guided Practice: Solving Systems of Linear Equations using Elimination

1. Use the system of equations to complete the following.

$$\begin{cases} 3m + 4k = 24 \\ 2m - 8k = -12 \end{cases}$$

- a. Solve the system using elimination.

$$m = \underline{\hspace{2cm}}$$

$$k = \underline{\hspace{2cm}}$$

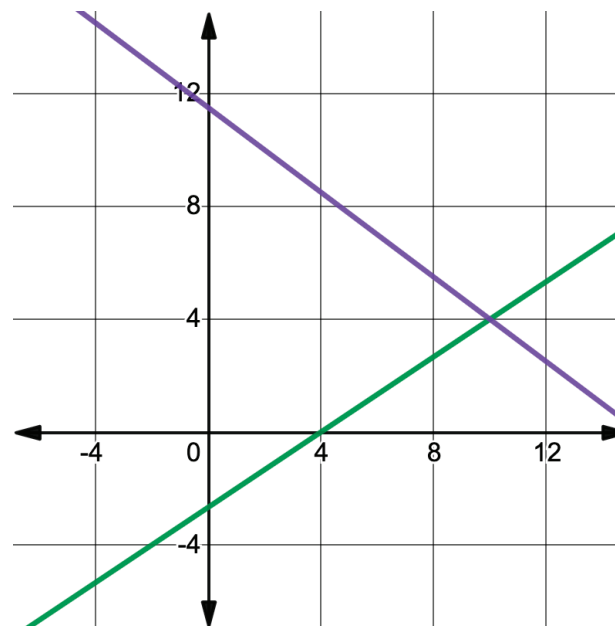
- b. Verify the solution algebraically using substitution.

2. Sometimes in order to solve a system by elimination, both equations must first be multiplied by different factors to eliminate one variable when adding or subtracting.

a. Discuss with your partner what factors could be multiplied by either equation in the system $\begin{cases} 2x - 3y = 8 \\ 3x + 4y = 46 \end{cases}$ to solve by elimination.

b. Solve the system using elimination.

c. The graph of the system is shown.



Discuss with your partner whether or not the graph verifies the solution found in Part B. If it does not, revisit Part B to find any errors.



5.9.4 Your Turn!

1. Solve the system of equations using elimination.

$$\begin{cases} 4x - y = -10 \\ -12x + 7y = 22 \end{cases}$$

2. Verify the solution algebraically.



5.9.5 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with solving systems of equations using elimination.



2. Explain why you chose that level of confidence.

Unit 5, Lesson 10: Solving Systems of Equations by Elimination and Interpreting Solutions



Benchmarks

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or nonviable options.



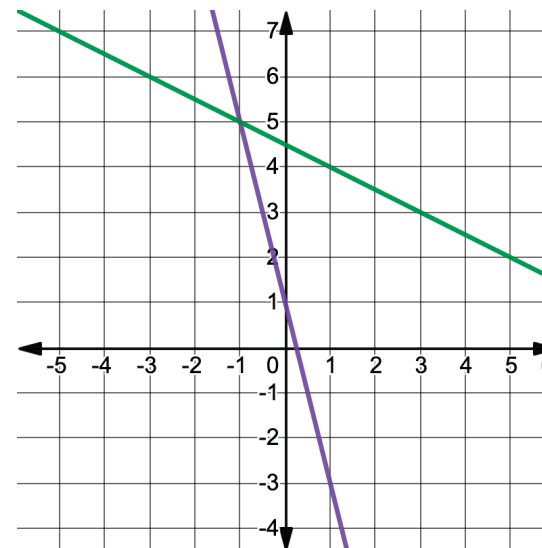
Learning Targets

- ☐ I can solve systems of equations by elimination.
- ☐ I can determine the constraints to a system of equations in a real-world context.
- ☐ I can interpret the solution to a system of equations in a real-world context.



5.10.1 Warm-Up: Bell Work

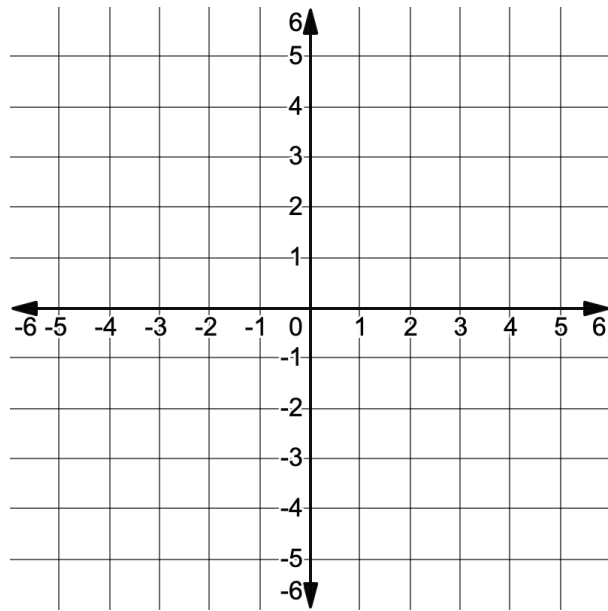
1. The system of equations $\begin{cases} 4x + y = 1 \\ x + 2y = 9 \end{cases}$ is graphed.



- a. Identify the solution to the system.
- b. Multiply the equation $4x + y = 1$ by each multiple indicated in the table to generate equivalent equations.

Multiple	Equivalent Equation
2	
-5	
$\frac{1}{3}$	

- c. Graph the three equivalent equations generated in Part B on the coordinate grid.



- d. What do you notice about the graphs of the equivalent equations?



5.10.2 Collaboration: Generating Equivalent Systems to Solve by Elimination

1. Chenille and Tobias are solving the system of equations
- $$\begin{cases} \frac{g}{2} + \frac{h}{8} = 4 \\ \frac{g}{3} - \frac{h}{2} = -2 \end{cases}$$
- using elimination.

Each of their first two steps are shown in the table.

Chenille's Work	Tobias' Work
$24\left(\frac{g}{2} + \frac{h}{8} = 4\right)$ $6\left(\frac{g}{3} - \frac{h}{2} = -2\right)$	$12\left(\frac{g}{2} + \frac{h}{8} = 4\right)$ $18\left(\frac{g}{3} - \frac{h}{2} = -2\right)$
$12g + 3h = 96$ $2g - 3h = -12$	$6g + \frac{3}{2}h = 48$ $6g - 9h = -36$

- a. Work with your partner to compare and contrast the methods used by each student.

How were their methods similar?	How were their methods different?

- b. Solve the systems of equations generated by Chenille and Tobias.

Chenille's System	Tobias' System
$12g + 3h = 96$ $2g - 3h = -12$	$6g + \frac{3}{2}h = 48$ $6g - 9h = -36$

- c. What do you notice about the solution to each equivalent system of equations?
- d. Verify algebraically whether or not the solution to Chenille and Tobias' systems satisfies the original system, $\begin{cases} \frac{g}{2} + \frac{h}{8} = 4 \\ \frac{g}{3} - \frac{h}{2} = -2 \end{cases}$.



5.10.3 Guided Practice: Solving Real-World Problems Represented by Systems of Equations

- The 2019-2020 NBA Season was Tyler Herro's rookie year with the Miami Heat. In game 1 of the Eastern Conference Finals, Herro scored 16 points against the Boston Celtics. He did not score any free throws during that game, but he scored 7 baskets, including two-point and three-point field goals. This situation can be represented by the system of equations, $\begin{cases} f + t = 7 \\ 2f + 3t = 16 \end{cases}$, where f is the number of two-point field goals made and t is the number of three-point field goals made.
 - Solve the system of equations using elimination.
 - Interpret the solution.
 - Describe the constraints on f and t using set-builder notation.

2. The Environmental Protection Agency, or EPA, tracks and records the amount of carbon dioxide (CO_2) emissions each year in the United States from different sectors of the U.S. economy.
- The total CO_2 emitted, m , in millions of metric tons, from the transportation sector is modeled by the equation $m = 9.8a + 1693.3$, where a is the number of years since 2008.
 - The electricity generation sector's total CO_2 emitted, m , in millions of metric tons is modeled by the equation $119a + 2m = 4634$, where a is the number of years since 2008.
- a. Javi suggests solving the system of equations $\begin{cases} m = 9.8a + 1693.3 \\ 119a + 2m = 4634 \end{cases}$ using elimination. Leah says she would prefer to solve it using substitution. Discuss with your partner which method you would prefer to use.
- b. Solve the system using your preferred method.
- c. Compare your work and solution with your partner.
- d. Interpret the solution.
- e. Discuss with your partner whether or not negative values for a or m are viable in this context.
- f. Describe the constraints on a and m using inequalities.





5.10.4 Your Turn!

1. Solve the system of equations using elimination.

$$\begin{cases} \frac{1}{3}x + \frac{4}{3}y = -5 \\ 6x + 8y = 16 \end{cases}$$

2. Zina runs an online store and uses pre-paid, flat-rate shipping boxes to send out orders. She bought a total of 12 boxes, some medium flat rate, m , and others large flat rate, g . Each medium box costs \$15.50 and each large box costs \$21.90. She spent a total of \$211.60. This situation can be represented by the system of equations shown.

$$\begin{cases} m + g = 12 \\ 15.5m + 21.9g = 211.6 \end{cases}$$

- a. Solve the system of equations.

- b. Interpret the solution.

- c. Describe the constraints on m and g in this context.



5.10.5 Wrap-Up: Error Analysis

1. Jeremiah solved the system of equations $\begin{cases} 0.15s + v = 6 \\ 0.03s - v = -4 \end{cases}$ using elimination. His work is shown.

$$\begin{array}{r} 0.15s + v = 6 \\ -(0.03s - v = -4) \\ \hline 0.12s = 10 \end{array}$$

$$s = 83\frac{1}{3}$$

$$\begin{array}{r} 0.03\left(83\frac{1}{3}\right) - v = -4 \\ 2.5 - v = -4 \\ v = 6.5 \end{array}$$

$$\left(83\frac{1}{3}, 6.5\right)$$

- Circle where Jeremiah made an error in his work.
- Find the correct solution to the system.

Unit 5, Lesson 11: Writing and Solving Systems of Equations for Real-World Contexts



Benchmarks

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can write and solve systems of equations in a real-world context.
- ☐ I can determine the constraints to a system of equations in a real-world context.
- ☐ I can interpret solutions as viable or non-viable in a context.



5.11.1 Warm-Up: Learning Strategies

1. Learning is not something that just happens, it's something you *do*.
 - a. Place a checkmark next to the actions you will choose to do today.
 - ☐ I will review my notes and my work.
 - ☐ I will ask questions when I need help.
 - ☐ I will set and maintain a high standard for myself.
 - ☐ I will ask for help if I'm not sure how to get started.
 - ☐ I will remember that my brain grows when I make mistakes.
 - b. Explain which, if any, strategy you picked demonstrates a growth mindset.



5.11.2 Guided Practice: Writing Systems of Equations

1. Scientists from NASA have focused their efforts to compare the amount of carbon dioxide (CO_2) emitted from different forms of travel.

Vehicle	CO_2 Emissions From Starting the Engine, in Kilograms (kg)	CO_2 Emissions per Mile Traveled, in kg
Car	1	0.25
Jet	2	0.14

- a. Write a system of equations to represent the total amount of emissions, C , from each vehicle from the time the engine is started when traveling a certain number of miles, m .
- b. Solve the system using the method of your choice.
- c. At what approximate distance will the jet and the car emit the same amount of CO_2 ?

- d. Shayla and Firenze discussed the constraints that exist in this situation.

Shayla said, "There are no constraints on m because the vehicles can travel any number of miles, including parts of miles. However, C must be positive because carbon dioxide is either emitted or not."

Firenze replied, "I agree with you about the carbon dioxide, but you also can't travel a negative number of miles, so the variable, m , cannot have any negative values either."

Represent the constraints in this context using inequalities.

- e. Firenze concluded from the context that jets are better for the environment because they emit less CO_2 . Discuss with your partner why this may not be an accurate interpretation.



5.11.3 Collaboration: Interpreting Solutions in Context

1. The equation $t = 71.03x + 3290.51$ represents the relationship between the total average cost of tuition each year, t , in dollars, for a full-time student attending a public university, and years since 2011, x . The total average cost of tuition for a full-time student attending a private university was \$17,710.25 in 2011, and has increased by an average of \$919.25 per year since then.
 - a. Write a second equation to represent the relationship between total average cost, t , and years since 2011, x , for private university tuition.
 - b. Using any method, solve the system of equations created with the two equations.
 - c. In what year did both private and public universities cost the same amount?

- d. What was the average total cost of tuition at that time?
2. Tyler prepares overnight oats for his breakfast each morning during the work week. He adds chia seeds for added nutrition. He buys the chia seeds and steel-cut oats he uses at a local bulk store. Last month, he purchased 0.8 pounds of chia seeds and 2.5 pounds of steel-cut oats for \$8.82. This month, he bought 0.4 pounds of chia seeds and 3 pounds of steel-cut oats for \$7.79.
- a. Write a system of equations that could be used to find the price per pound for chia seeds and steel-cut oats at the bulk store. Define the variables used.
- b. Solve the system of equations. Round your answers to the nearest thousandth.
- c. Discuss with your partner any constraints on the variables in this context. Then, summarize your discussion.
- d. Based on your work in Part B and discussion in Part C, interpret a viable solution to the problem.



5.11.4 Your Turn!

1. The average homeowner in Florida pays \$150 per month on their energy bills. Some homeowners choose to install solar panels to reduce their monthly energy costs to an average of \$55 per month. The cost of purchasing and installing solar panels is \$13,250 on average.
 - a. Write a system of equations that can be used to determine how many months it would take for the total cost of energy bills without solar panels to cost the same as an energy bill with solar panels. Define the variables used.
 - b. Solve the system of equations. Round your answers to the nearest thousandth.
 - c. Interpret the solution.



5.11.5 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate how you feel about each learning target.
 - a. I can write and solve systems of equations in a real-world context.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
 - b. I can interpret solutions as viable or non-viable in a given context.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

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